

BST Working Paper — Volume 2: The Framework — Chapter 2: Standard Model

2026-05-19 (Tuesday volume:chapter reorganization)

Section 5: The Fine Structure Constant

Rewritten 2026-09-13 (Lyra, Round 143 L2; Keeper audit pending) as a pointer into the Spine plus the referee apparatus. The May 2026 text is kept below under its own heading as the record; the 2026-09-11 head-note that stood here is folded into “Where it stands.” Facts are from the register (K1813, K1826, K1872, K1873, K1875, K1884, T186, T1136, T1448) and not from the May text. Last accuracy-synced: 2026-09-13 / K1895.

5.0 The question

Is 137 in the geometry?

It is the first question anyone asks of a program like this one, and the right one. The fine-structure constant is dimensionless, measured to about one part in ten billion, and taken by the Standard Model as an input. A geometry that produced it would be doing something no theory has done. This section says what we found, what we proved cannot work, and what a referee can check with a calculator. The full treatment is Lecture 8 of the Spine (Curriculum/Spine_DIV5_QM_GR_SM/Lecture_08_Alpha_the_Honest_Chapter.md); what follows is the pointer and the apparatus.

5.1 Where it stands

Three things are simply true. The measured value is $\alpha^{-1} = 137.035999 \dots$. The shape has a computed invariant $N_{\max} = N_c^3 n_C + \text{rank} = 27 \cdot 5 + 2 = 137$, an integer combination of numbers fixed in March 2026 before any comparison (T186); inside the program it is a channel count and it enters the definition of the clock (Lecture 4). And the two agree to three significant figures and disagree at the fourth. Everything else is about the word that goes between them, and the word is identified. In August we ran the null our own reviewer paper demands: form every allowed small-integer expression of the shape and count how many measured constants land as well as 137 does. Twenty-three to twenty-six of seventy-six targets scored at least as well;

the real attractors of that integer soup are $128 = 2^8$ and $125 = n_C^3$, and 137 sits in the bulk (K1813). The same instrument, the same day, gave the muon-mass form fewer than one chance in sixty of being an accident (K1814) — the instrument discriminates, and on this number it says coincidence. Since 2026-08-11 the register has carried the instruction, in those words, not to cite α externally as derived. The three “derivations of $N = 137$ ” in the May record below are three identifications, and the record is kept because the attempt was serious.

The route that would have been a proof is the one Wyler tried in 1969: read α off a ratio of volumes on this very family of domains. We tested it the way it should have been tested in 1971. The unique *forced* reading — the Born-normalised bare vertex on the Šilov boundary, its power fixed, the candidate landings and their tolerances written down and hashed before the number — computes to $8\pi^3/3 \approx 82.68$, the Hua volume of the Lie sphere, not 137 (K1826, toy 5496; Landing C under the pre-registered protocol). That closes the whole class by pincer: the forced member is refuted by computation, and the free family — every reading that leaves a choice to the reader, Wyler’s included — had already been shown to fit anything. Fired and lost, certified. Two more identifications sit beside it and are named as such: the two-loop coefficient of the electron’s anomalous moment, which a trace on the geometry reproduces term for term — because the four terms are Petermann and Sommerfield’s 1957 closed form (K1872; T1448 re-tiered identified) — and the six-flavour one-loop coefficient $b_0 = 7$ read as g (K1875). In 137, where it appears, is a normalisation unit and not a prediction (K1873). One claim was withdrawn the day it was made: the program’s clock defines its α through the success of a write, so it is constant by construction and no laboratory drift bound reaches it (T1136, K1884). What is still open is exactly Wyler’s question with the volume reading removed: if α is in this geometry, it is not in its volume, and the clock’s reading of it as a rate has a definition but not a derivation.

5.2 Apparatus — the arithmetic a referee will want, labelled as apparatus

Nothing in this subsection is a claim. It is the computation behind the sentences above, kept so that the reader who came to find the flaw does not have to reconstruct it.

Wyler’s formula, as arithmetic. Hua’s volume of the Lie ball in five dimensions is $\text{Vol}(D_{IV}^5) = \pi^5/(2^4 \cdot 5!) = \pi^5/1920 = 0.1593853 \dots$ (Elie’s toy 5752 finds 0.1570 by uniform sampling, 1.5σ from the closed form). Wyler’s expression is

$$\alpha_W = \frac{9}{8\pi^4} \left(\frac{\pi^5}{2^4 \cdot 5!} \right)^{1/4} = 0.00729734813 \dots, \quad \alpha_W^{-1} = 137.036082,$$

against CODATA 137.035999: a relative difference of 6×10^{-7} . That is the number Wyler had, and it is real. What it is *not* is forced: the exponent 1/4, the factor 9 and the $8\pi^4$ are each one choice among several the geometry permits, and the pre-registered null of K1813 is the measurement of how much such freedom buys. The

May record’s “9 from the Weyl vector” is correct group theory — the restricted root system of $SO_0(5, 2)$ is B_2 with long roots $e_1 \pm e_2$ of multiplicity 1 and short roots e_1, e_2 of multiplicity $n_C - 2 = 3$, so $\rho = (5/2, 3/2)$ and $9 = (2\rho_2)^2$ — and the step from that to “ $\alpha = \rho_2^2/(2\pi^4) \times \text{Vol}^{1/4}$ ” is a reading chosen after the target was known. Robertson’s 1971 critique named the same worry (the radius set to one; a Poisson-kernel coefficient) and Gilmore’s 1972 note the scaling; the citations are pinned in Lecture 8’s “Where to look.”

The forced vertex. The only reading with no choice left in it — the Born-normalised bare vertex on $\check{S}$ — is the Hua volume of the Lie sphere, $2\pi^{n/2+1}/\Gamma(n/2)$ at $n = 5$, which is $8\pi^3/3 = 82.683$. The protocol (landings A/B/C and their tolerances, hashed before the run) and the computation are K1826 and `play/toy_5496_*`. The three regroupings of Wyler’s number in the May record (Section 5.5’s $9/8 \times \pi^{-4} \times 0.632$) are algebraically identical to α_W and add no evidence.

The anomalous moment. The four-term trace of T1448 equals, term for term, Petermann–Sommerfield’s $C_2 = 197/144 + \pi^2/12 - (\pi^2/2)\ln 2 + (3/4)\zeta(3) = -0.32848$; K1872 is the comparison. (Lecture 1’s $C_2 = 6$ is a different object under the same letter; the QED coefficient is meant here.)

What the May record gets wrong as mathematics, so you need not check it twice. Its Section 5.3 writes $D_{IV}^5 = SO(7)/[SO(5) \times SO(2)]$; that is the compact dual (the quadric Q^5), and the domain is $SO_0(5, 2)/[SO(5) \times SO(2)]$. Its “CR dimension $N_c + N_w = 5$ forces the domain” and its Cartan table are the retired uniqueness argument; the surviving selection is Lecture 2’s — the characteristic multiplicity $a = 3$ selects D_{IV}^5 alone, with the identification of a with the colour count as the program’s one input to that choice. Its rigidity argument (Section 5.3) is true of any invariant of a fixed domain and says nothing about whether α is one. Its cost function (Section 5.4) tunes κ to the geometry after the fact and its Shannon decomposition (Section 5.5) regroupes Wyler’s factors; both are identifications by the test of K1813.

5.3 Tier line

- Measured: $\alpha^{-1} = 137.036$.
- Computed invariant: $N_{\max} = 137$.
- Identified — do not cite externally as derived: every expression for α^{-1} in this program, the forward form included; the a_e coefficient; $b_0 = 7$.
- Fired and lost, certified: the volume reading — forced vertex $8\pi^3/3$.
- Withdrawn: α drift.
- Open: whether α is in the geometry at all.

What would make this section wrong: nothing here is exposed to a laboratory number. It would become wrong in the other direction if someone derives α by a route that is neither a volume reading nor a forward form, target hidden, protocol hashed. The instruments to check such a claim are in the repository.

5.4 May 2026 record

The text below is the May 2026 narrative, kept as the record of the attempt and unedited except for the two annotations already in it (2026-09-11). Read it with Sections 5.1–5.3 in hand: its “derivations” are identifications, its 0.0001% is the fit of a chosen reading, and its Section 5.3 names the compact dual where it means the domain.

May 5.1 — Wyler’s Formula In 1969, Armand Wyler computed a geometric ratio on D_{IV}^5 and obtained $\alpha = 1/137.036$, matching the measured fine structure constant to the available precision. His paper was published in Comptes Rendus but widely dismissed because he provided no physical reason why D_{IV}^5 should be the relevant domain.

Robertson (1971) asked the critical question: “Why this domain?” Without a physical motivation, Wyler’s result was numerology — a correct number from an unjustified calculation.

BST answers Robertson’s question. D_{IV}^5 is the configuration space of the BST contact graph, derived from the substrate geometry $S^2 \times S^1$ through established theorems in CR geometry and Lie group theory. The physical motivation is the substrate itself.

The explicit formula. The Bergman kernel of D_{IV}^5 evaluated at the Shilov boundary $S^4 \times S^1$ gives a natural metric in which the ratio of scales is set by the domain volume. The Wyler formula is:

$$\alpha = \frac{9}{8\pi^4} \left(\frac{\pi^5}{2^4 \cdot 5!} \right)^{1/4} = \frac{9}{8\pi^4} \left(\frac{\pi^5}{1920} \right)^{1/4}$$

The exponent 1/4 arises from the normalization of the Bergman kernel on D_{IV}^5 : the Plancherel measure for spherical functions on the symmetric space $SO_0(5, 2)/(SO(5) \times SO(2))$ introduces $\text{Vol}(D_{IV}^5)^{1/4}$ as the natural geometric scale for the principal series representation associated with the S^1 winding mode. In the Harish-Chandra construction, the c-function for the relevant representation contributes a $V_5^{1/4}$ factor from the boundary-to-bulk normalization of the Bergman kernel — a result of the Bergman metric transformation law, not a simple ratio of dimensions. The volume $\text{Vol}(D_{IV}^5) = \pi^5/1920$ is fixed by the group theory of $SO_0(5, 2)/(SO(5) \times SO(2))$. The full derivation via the Harish-Chandra c-function follows immediately below.

Geometric reading of the “9”: Harish-Chandra Weyl vector. The numerator 9 in the Wyler formula is not an empirical constant. It is determined by the Weyl vector of the restricted root system of $SO_0(5, 2)$.

The symmetric space $D_{IV}^5 = SO_0(5, 2)/(SO(5) \times SO(2))$ has rank 2, with restricted root system of type B_2 . The positive roots and their multiplicities are:

Root	Multiplicity
$e_1 - e_2$ (long)	1
$e_1 + e_2$ (long)	1
e_1 (short)	$n_C - 2 = 3$
e_2 (short)	$n_C - 2 = 3$

The Weyl vector $\rho = \frac{1}{2} \sum_{\alpha > 0} m_\alpha \alpha$ has components:

$$\rho = \left(\frac{n_C + q - 2}{2}, \frac{n_C - q}{2} \right) = \left(\frac{5}{2}, \frac{3}{2} \right) \quad (q = 2)$$

The S^1 winding direction of the Shilov boundary $\Sigma = S^4 \times S^1$ is governed by $\rho_2 = (n_C - q)/2 = 3/2$. The fine structure constant is the coupling of the minimal S^1 winding circuit to the Bergman vacuum, given by the spectral weight of the S^1 mode:

$$\alpha = \frac{\rho_2^2}{2\pi^{2q}} \times (\text{Vol}(D_{IV}^{n_C}))^{1/4} = \frac{(n_C - 2)^2}{8\pi^4} \times \left(\frac{\pi^5}{1920} \right)^{1/4}$$

For $n_C = 5, q = 2$: $\rho_2^2 = (3/2)^2 = 9/4$, so $\rho_2^2/(2\pi^4) = 9/(8\pi^4)$, recovering the Wyler prefactor exactly. The numerator $9 = (n_C - 2)^2 = \rho_2^2 \times 4$ — twice the second Weyl vector component, squared. The denominator $8\pi^4 = 2\pi^{2q}$ — the π^{2q} factor from the $\text{SO}(q) = \text{SO}(2)$ fiber normalization.

Numerical verification (computed from first principles, March 2026):

Quantity	Value
$\text{Vol}(D_{IV}^5) = \pi^5/1920$	0.159385252...
α_{Wyler}	0.007297348...
$1/\alpha_{\text{Wyler}}$	137.036082
$1/\alpha_{\text{observed}}$ (CODATA 2018)	137.035999
Precision	0.0001%

This is not a fit. No parameters are adjusted. The formula outputs the observed fine structure constant from the volume of a bounded symmetric domain that BST identifies on independent geometric grounds as the configuration space of the substrate. The agreement to 0.0001% — six significant figures — from a formula whose only input is a group-theoretic volume is the most precise parameter-free prediction in the BST framework.

May 5.2 — The Packing Number The fine structure constant $\alpha^{-1} = 137$ is the channel capacity of the S^1 fiber — the maximum number of non-overlapping circuits. This is a topological packing number determined by the geometry of the Shilov boundary of D_{IV}^5 .

The number 137 is a Euclidean prime expressible as a sum of two squares: $137 = 4^2 + 11^2$. This decomposition is not incidental — the two squared terms relate to the two packing dimensions of the domain.

May 5.3 — Topological Rigidity of $\alpha = 1/137$ Status: The Casimir stability conjecture is superseded. The stability of $\alpha = 1/137$ is topological, not dynamical. The monotone Casimir result (confirmed by computation) is the expected signature of this.

The Casimir computation. A full Seeley-DeWitt regularization of $\zeta_{S^4 \times S^1}^{\text{ren}}(-1/2; \rho)$ was carried out (March 2026), splitting the spectral zeta into UV and finite pieces via Poisson resummation of the S^1 modes:

$$\zeta^{\text{ren}}(-1/2, \rho) = \underbrace{C_{UV} \cdot \rho}_{\text{UV piece}} + \underbrace{-\rho \sum_{n=1}^{\infty} I_n(\rho)}_{\text{winding modes}}$$

where $C_{UV} = \frac{1-\gamma_E}{2} \zeta_{S^4}(-1) = +0.003207$ and $I_n(\rho) \sim e^{-4\pi n \rho}$. At $\rho = 137$: $I_1(137) \sim 10^{-748}$. Both pieces are monotone. The flat-product Casimir energy has no minimum at $\rho = 137$. The Bergman boundary term is also monotone. No Casimir mechanism — flat or curved — selects $\rho = 137$ as an energy minimum. This is the correct result. The monotonic Casimir is not a failure of the theory; it is evidence that the stability mechanism is topological.

The topological rigidity argument. The derivation chain is fully determined by discrete data:

1. The BST contact structure is a strictly pseudoconvex CR manifold with CR dimension 5 — fixed by the gauge structure $N_c + N_w = 3 + 2 = 5$ (three quark colors, two electroweak dimensions).
2. The isotropy group is $\text{SO}(5) \times \text{SO}(2)$ — confirmed numerically (Section 4, seven independent checks all pass).
3. By Cartan's classification theorem, the Hermitian symmetric space with this group and isotropy is the unique type-IV domain in 5 complex dimensions: $D_{IV}^5 = \text{SO}(7)/[\text{SO}(5) \times \text{SO}(2)]$.

Each step is forced. The Cartan classification is a theorem — given the group and isotropy, the domain is unique. The classification is discrete:

Type	Domain	Why not BST
I	$SU(p,q)/S[U(p)\times U(q)]$	Different isotropy group
II	$SO^*(2n)/U(n)$	Different isotropy group
III	$Sp(n,\mathbb{R})/U(n)$	Different isotropy group
IV	$SO(n,2)/[SO(n)\times SO(2)]$	$n = 5$ from CR dimension
V	E_6 quotient	Different isotropy group
VI	E_7 quotient	Different isotropy group

To change the domain type requires changing the isotropy group, which requires changing the CR dimension, which requires changing $N_c + N_w$, which requires a different number of quark colors or electroweak dimensions. None of these can change continuously. They are discrete.

Consequence for α . The Wyler formula (Section 5.1) gives α as a geometric invariant of D_{IV}^5 — a ratio of canonical Bergman volumes. This invariant is not a parameter to be minimized over. It is as fixed as π is for a circle. There is no continuous family of domains parameterized by ρ between which the system could slide. D_{IV}^5 is what it is; $\alpha = 1/137.036$ is what that domain implies.

Stability. The vacuum cannot decay to a different value of α because there is no continuous path to a configuration with different α . A different α would require a different ρ , which would require a different domain type, which is a discrete jump — not a continuous deformation. Tunneling requires a continuous path through configuration space. No such path exists between Cartan types.

This is stronger than any energy minimum. A Casimir minimum can in principle be tunneled through if the barrier is finite. A topological classification cannot be tunneled through because there is no barrier — there is no path at all. The stability of $\alpha = 1/137$ is the same kind of stability as the stability of a winding number: not enforced by a potential, but by the absence of any continuous deformation that could change it.

The correct description is Riemannian rigidity: D_{IV}^5 is an irreducible Hermitian symmetric space of non-compact type, and its geometry is completely determined (up to overall scale) by its type in the Cartan classification. The dimensionless ratio ρ has no moduli. Perturbations that would change ρ would break the $SO(7)/[SO(5) \times SO(2)]$ symmetry — taking the substrate out of the Cartan classification entirely. The Casimir energy need not select $\rho = 137$ because the geometry already did, and topology locks it in.

May 5.4 — A Second Independent Derivation: The Substrate Cost Function The Wyler formula reads α^{-1} off a volume ratio on D_{IV}^5 (a reading whose forced version computes to $8\pi^3/3$, not 137 — K1826; see the head of this section). A completely different geometric construction — the cost function of the self-maintaining substrate — selects the same integer $N = 137$ independently.

The cost function. A substrate with channel capacity ρ incurs two competing costs: a geometric cost proportional to ρ (more slots require more structure to maintain) and a computational cost that decreases with ρ (a larger channel has more room for error correction, with Shannon capacity $\sim \ln(\rho + 1)$ per slot):

$$C(\rho) = \underbrace{\rho}_{\text{geometric}} + \underbrace{\frac{\kappa}{(\rho + 1) \ln(\rho + 1)}}_{\text{computational}}$$

The parameter κ is the ratio of computational to geometric cost — how many slot-units of geometric cost the substrate pays per unit of computational requirement. Setting $dC/d\rho = 0$ places the minimum at:

$$\kappa = \frac{(\rho + 1)^2 \ln^2(\rho + 1)}{1 + \ln(\rho + 1)}$$

For $\rho^* = 137$: $\kappa_{\text{exact}} = 78,004$.

Geometric identification of κ . The S^4 spherical harmonic degeneracy at degree l satisfies the exact identity:

$$d_l - d_{l-1} = (l + 1)^2 \quad (\text{provable from } d_l = (l + 1)(l + 2)(2l + 3)/6)$$

At the resonant degree $l^* = \dim_{\mathbb{C}}(D_{IV}^5) = 5$, the spectral step is $d_5 - d_4 = 6^2 = 36$. The Bergman curvature of D_{IV}^5 mixes adjacent harmonic degrees at the resonant level with mixing weight:

$$w = \frac{1}{l^*(l^* + 1)^2} = \frac{1}{5 \times 36} = \frac{1}{180}$$

The Bergman-weighted mode density at degree l^* is $d_{\text{eff}} = (w \cdot d_4 + d_5)/(1 + w) = 90.801$. The geometric identification of κ is:

$$\kappa = \frac{N_{\text{max}} \cdot d_{\text{eff}}}{\text{Vol}(D_{IV}^5)} = \frac{137 \times 90.801 \times 1920}{\pi^5} = 78,047$$

Every factor is fixed by D_{IV}^5 geometry — no free parameters. With this κ , the continuous minimum falls at $\rho^* = 137.035$, and $\lfloor \rho^* \rfloor = 137$.

Three readings that land on $N = 137$ (identifications, all three — see the note at the head of this section; the May text called them derivations):

Method	Geometric input	Continuous value	Physical N
Wyler formula	Volume ratio D_{IV}^5 /Shilov boundary	137.036	137
Cost function (Bergman- corrected)	Mode density at $\dim_{\mathbb{C}}$ / domain volume	137.035	137
Cost function (first-order)	Same, leading correction $-1/l^*$	137.034	137

The Wyler formula and the Bergman-corrected cost function agree to 5 parts per million — from completely different geometric calculations on D_{IV}^5 . The Wyler formula uses integrated volumes; the cost function uses the harmonic spectrum at a specific degree and the Bergman curvature. Their agreement at the level of 5 ppm is a non-trivial internal consistency check on the theory. Both give $[\rho^*] = 137$.

Full derivation: `notes/BST_CostFunction_Kappa.md`.

May 5.5 — Shannon Interpretation: Alpha as Optimal Code Rate The Wyler formula and the cost function each land on 137 from Bergman geometry — two identifications, not derivations (see the head of this section). A third perspective reveals the same number from Shannon information theory, providing a physical interpretation: α is the fraction of the substrate's channel capacity that carries signal; the remaining 136/137 is error correction overhead.

Von Mises-Packing Equivalence. On $S^2 \times S^1$, for small concentration κ :

$$\frac{1}{N_{\text{pack}}} = C_{\text{vonMises}} = \frac{\kappa^2}{4} = \alpha$$

where $\kappa = 2/\sqrt{137}$ is simultaneously the sphere packing footprint radius and the von Mises phase channel noise concentration. Packing (geometry) equals capacity (information) through a single parameter.

Three-factor decomposition. Wyler's formula decomposes into three independently derivable factors:

$$\alpha = \underbrace{\frac{N_c^2}{2^{N_c}}}_{9/8 \text{ (color rate)}} \times \underbrace{\frac{1}{\pi^{n_C-1}}}_{1/\pi^4 \text{ (curvature penalty)}} \times \underbrace{\left[\frac{\pi^{n_C}}{1920} \right]^{1/(n_C-1)}}_{0.632 \text{ (volume reach)}}$$

The bandwidth killer is curvature: $1/\pi^4 \approx 1\%$. The S^2 boundary eats 99% of the channel capacity. This is why α is small — not because of some mysterious fine-tuning, but because curved geometry is expensive.

Signal/curvature/noise $\rightarrow$ force identification. The three factors in α map directly to the three geometric layers of Section 14:

Factor	Role in α	Force sector	Scale
$N_c^2/2^{N_c} = 9/8$	Signal (color rate)	Strong force	$\sim 1 \text{ GeV}$
$1/\pi^{n_c-1} = 1/\pi^4$	Curvature penalty	Weak force	$\sim 100 \text{ GeV}$
$(\pi^{n_c}/1920)^{1/(n_c-1)}$	Volume reach (noise floor)	Dark matter	$\sim \text{galactic}$

The curvature penalty $1/\pi^4$ is why the weak scale is $\sim 100\times$ the strong scale: the S^2 boundary geometry imposes a factor of $\pi^4 \approx 97$ between the two sectors. The ratio $m_W/m_p = n_c/(8\alpha)$ contains the full $1/\alpha$ factor, which itself contains the curvature penalty — the weak force IS the curvature cost of the substrate. The noise floor factor sets the dark matter acceleration scale a_0 where channel noise begins to dominate gravitational signal.

1920 as coding symmetry. $1920 = |S_5 \times (\mathbb{Z}_2)^4| = 5! \times 2^4$: the number of permutations of 5 phase channels ($5! = 120$) times the number of relative phase signs ($2^4 = 16$, with 4 of 5 independent). This equals $|W(D_5)|$, the Weyl group of the D_5 root system ($= \text{SO}(10)$, the GUT group). The Bergman volume $\text{Vol}(D_{IV}^5) = \pi^5/1920$ is a coding quantity: the number of distinguishable codewords in a 5-phase code with natural symmetry.

Alpha running as dimensional flow. At energy Q , the effective curvature dimensionality $d_{\text{eff}}(Q)$ decreases as more boundary modes are resolved:

$$\alpha(Q) = \frac{N_c^2}{2^{N_c}} \cdot \frac{1}{\pi^{d_{\text{eff}}(Q)}} \cdot \left[\frac{\pi^{n_c}}{1920} \right]^{1/(n_c-1)}$$

d_{eff} decreases from 4.00 at m_e to 3.94 at m_Z . A change of only 1.5% in boundary dimensionality accounts for the entire running of α from 1/137 to 1/128, matching standard QED to 0.5%. QED and QCD run in opposite directions because they have opposite noise sources: EM noise from the S^2 boundary (decreases at short distance) vs. QCD noise from the D_{IV}^5 bulk (also decreases, but bulk mode decoupling reduces confinement).

Bergman-Fisher duality. At the origin of D_{IV}^5 :

$$g_B(0) = \frac{n_C}{\alpha} \cdot g_F(\kappa_\alpha)$$

The Bergman metric equals the Fisher information metric times the number of modes. This exact identity proves that the Bergman kernel and Shannon capacity measure the same information at different scales.

Full derivations: `notes/BST_Shannon_Alpha_Paper.md`, `notes/BST_Shannon_Alpha_Theorem.py`, `notes/BST_AlphaRunning_Shannon.py`, `notes/BST_BergmanFisher_Theorem.py`.

Section 6: Structured Unification

6.1 The Standard Model’s “Failure” to Unify

Standard grand unification predicts that the three gauge couplings converge to a single value at the GUT scale. They do not: the measured couplings, extrapolated via renormalization group flow, miss the convergence point. This “failure to unify” has been a persistent problem for four decades.

6.2 BST’s Structured Unification

BST reinterprets this as a success. The three force sectors correspond to three packing dimensions on S^1 within D_{IV}^5 .

Electroweak sector: The electromagnetic and weak couplings share two packing dimensions. They unify at $N_{GUT} = (2\pi)^2 = 4\pi^2 \approx 39.48$. The measured SU(5) unification coupling is ~ 40 , within 1.3%.

Strong sector: The strong coupling occupies a third packing dimension with topological constraint from the Z_3 center of SU(3). This gives $\alpha_3(M_{GUT}) = 4\pi^2/N_c = 4\pi^2/3 \approx 13.16$.

The couplings do not converge to a point. They converge to a structure: electroweak unification at $4\pi^2$, strong coupling at $4\pi^2/N_c$. The Standard Model was telling us the answer for forty years. The “failure” was a misinterpretation of what unification means in a topologically structured framework.

6.3 The Weinberg Angle

The Weinberg angle $\sin^2 \theta_W$ measures the mixing between $SU(2)_L$ and $U(1)_Y$ in the electroweak sector. In BST, this mixing is geometric: it is the ratio of the color sector dimension to the total gauge-active dimension of D_{IV}^5 .

$$\sin^2 \theta_W = \frac{N_c}{N_c + 2n_C} = \frac{3}{3 + 10} = \frac{3}{13} = 0.23077$$

matching the $\overline{MS}$ value 0.23122 to 0.2% with no free parameters. The numerator $N_c = 3$ counts color directions; the denominator $N_c + 2n_C = 13$ is color ($N_c = 3$) plus the real dimension of D_{IV}^5 ($2n_C = 10$) — the total number of gauge-active real dimensions. The physical interpretation: $\sin^2 \theta_W$ measures what fraction of the gauge interaction comes from the color sector (hypercharge) versus the full geometric structure.

Consequences. The double angle gives $\cos 2\theta_W = 7/13$, connecting the Weinberg angle to the same integer $7 = n_C + 2 = p + q$ — the dimension of the defining

representation of $SO(5, 2)$ (signature $(5, 2)$), which this chapter called the genus in May; the genus of D_{IV}^5 is 5 (pinned 2026-09-09) — that appears in $\alpha_s = 7/20$, $H_{YM} = 7/(10\pi)$, and $\beta_0 = 7$. The W mass follows from the tree-level relation:

$$m_W = m_Z \sqrt{1 - 3/13} = m_Z \sqrt{10/13} = 79.977 \text{ GeV} \quad (0.5\% \text{ from observed } 80.377 \text{ GeV})$$

The 0.5% deviation is consistent with radiative corrections (loop effects shift m_W upward by ~ 0.3 – 0.7 GeV from tree level).

Relation to the standard GUT value $3/8$. The standard $SU(5)$ prediction is $\sin^2 \theta_W = 3/8 = 0.375$ at the GUT scale. In BST, $3/8$ arises from using the complex dimension n_C in the denominator instead of the real dimension $2n_C$: $N_C/(N_C + n_C) = 3/8$. The BST result $3/13$ is the low-energy value; the D_{IV}^5 geometry already incorporates the renormalization group flow. No explicit running is needed because the domain geometry encodes the full scale dependence.

Full derivation: `notes/BST_WeinbergAngle_Sin2ThetaW.md`.

6.4 Number of Colors

The number of quark colors $N_C = 3$ is the short-root multiplicity of D_{IV}^5 — a forced geometric integer. The Z_3 center (the thirds) follows *from* $N_C = 3$ and the topological closure on quark triads; the $SU(3)$ gauge *group* whose center this is imported (#108), but the number 3 and the Z_3 structure are geometric. Three quarks cycling through color orderings on $\mathbb{CP}^2$ require Z_3 closure — the circuit must return to its starting configuration after three steps. $N_C = 3$ is a geometric necessity, not a parameter.

6.5 The Electroweak Algebra as an Exact Isotropy Subalgebra

The electroweak gauge algebra $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$ is not merely consistent with D_{IV}^5 — it sits inside the isotropy algebra as an exact subalgebra.

The isotropy algebra of D_{IV}^5 is $\mathfrak{k} = \mathfrak{so}(5) \oplus \mathfrak{so}(2)$. The $\mathfrak{so}(2)$ factor is identified with $\mathfrak{u}(1)_{EM}$ (the complex structure generator of the S^1 fiber). Inside $\mathfrak{so}(5)$, using 5×5 antisymmetric matrix generators K_{ij} ($(K_{ij})_{ab} = \delta_{ia}\delta_{jb} - \delta_{ib}\delta_{ja}$), the following four generators:

$$T_1 = K_{02} + K_{13}, \quad T_2 = K_{03} - K_{12}, \quad T_3 = K_{01} - K_{23}, \quad Y = K_{01} + K_{23}$$

satisfy exactly the $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ commutation relations:

$$[T_1, T_2] = 2T_3, \quad [T_1, T_3] = -2T_2, \quad [T_2, T_3] = 2T_1, \quad [Y, T_i] = 0$$

verified numerically to machine precision. Therefore:

$$\mathfrak{k} = \mathfrak{so}(5) \oplus \mathfrak{so}(2) \supset \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y \oplus \mathfrak{u}(1)_{EM}$$

The complete electroweak gauge algebra sits inside the isotropy algebra of D_{IV}^5 as an exact subalgebra of codimension 6. This is not a consistency check — it is a theorem about the geometry of the domain.

Physical hypercharge — unique identification. The generator $Y = K_{01} + K_{23}$ found above involves color index $\{2, 3\}$ and does not commute with all of $SU(3)_c$. To find the physical hypercharge, one must search the full $\mathfrak{so}(5, 2)$ algebra using 7×7 generators J_{AB} (indices $\{0, 1, 2, 3, 4, 5, 6\}$, metric $g = \text{diag}(+1, +1, +1, +1, +1, -1, -1)$).

Among all 21 generators of $\mathfrak{so}(5, 2)$, exactly six are color-singlet (commute with the $SO(3)$ color generators J_{23}, J_{24}, J_{34}): the set $\{J_{01}, J_{56}, J_{05}, J_{06}, J_{15}, J_{16}\}$. These close to form the algebra $\mathfrak{so}(2, 2) \cong \mathfrak{sl}(2, \mathbb{R}) \oplus \mathfrak{sl}(2, \mathbb{R})$. Among these six, only one commutes with all three $SU(2)_L$ generators T_1, T_2, T_3 — verified numerically:

$$Y_{\text{phys}} = J_{56}$$

This is the $\mathfrak{so}(2)$ generator of the isotropy algebra $\mathfrak{k} = \mathfrak{so}(5) \oplus \mathfrak{so}(2)$ — the rotation in the S^1 fiber (indices $\{5, 6\}$ carry the two “timelike” directions of the $(5, 2)$ signature). Hypercharge is S^1 fiber winding. The identification is a theorem: no other generator in $\mathfrak{so}(5, 2)$ can serve as physical hypercharge.

Weinberg angle from the full $\mathfrak{so}(5, 2)$ geometry. Using the Killing form $B(X, Y) = 5 \text{Tr}_7(XY)$ on the full algebra:

- $T_3 = J_{01} - J_{23}$: combination of two elementary generators, $|B(T_3, T_3)| = 20$
- $Y_{\text{phys}} = J_{56}$: single elementary generator, $|B(J_{56}, J_{56})| = 10$

$$\sin^2 \theta_W|_{\mathfrak{so}(5, 2)} = \frac{10}{20 + 10} = \frac{1}{3}$$

This result is representation-independent (verified in both the 7×7 fundamental and the 21×21 adjoint). The factor of 2 between $|T_3|^2$ and $|Y_{\text{phys}}|^2$ has a transparent geometric origin: T_3 is a combination of two elementary rotations (the Hopf rotation J_{01} and the color-isospin rotation J_{23}), while Y_{phys} is a single elementary rotation (the S^1 fiber J_{56}).

From two-loop Standard Model renormalization group running, $\sin^2 \theta_W = 1/3$ corresponds to an energy scale $\mu \approx 10^{12}$ GeV — an intermediate scale between the electroweak scale and the GUT scale. This is distinct from both the $SU(5)$ GUT prediction ($3/8$ at $\sim 10^{16}$ GeV) and from $\sin^2 \theta_W = 1/2$ at M_{pl} (the value from $\mathfrak{so}(5)$ alone, before accounting for the fiber structure). BST predicts a specific intermediate unification scale from the geometry of D_{IV}^5 .

Complete Standard Model gauge group from D_{IV}^5 .

Factor	Generator(s)	Geometric origin
$SU(3)_c$	$J_{23}, J_{24}, J_{34} + 5 \mathbb{C}P^2$ holonomy generators	Holonomy of $\mathbb{C}P^2 \subset S^4$ (Shilov boundary)
$SU(2)_L$	$T_1 = J_{02} + J_{13}, T_2 =$ $J_{03} - J_{12}, T_3 = J_{01} - J_{23}$	Hopf fibration $S^3 \rightarrow S^2$ inside S^4
$U(1)_Y$	J_{56}	S^1 fiber rotation (isotropy $\mathfrak{so}(2)$)

The electroweak factors $SU(2)_L \times U(1)_Y$ are read off from the geometry — they are the spinor/isotropy structure of D_{IV}^5 (the reality-type census: the complex charge circle gives $U(1)_Y$, the quaternionic spinor gives $SU(2)_L$; both *derived*). The $SU(3)_c$ color factor is different, and the difference is the honest result: the geometry forces the number $N_c = 3$, the $\mathbb{C}^3$ color space, and the Z_3 center (the thirds), but $SU(3)$ as a gauge group is imported, not read off — color must be an *internal* symmetry (Coleman–Mandula), and the geometry’s internal commutant contains no color group (it is $U(1) \times SU(2) \times Z_2$). So two factors are *derived* and one factor’s group is *imported* on a *derived container* ($N_c = 3, \mathbb{C}^3, Z_3$) — and that split, not a shared geometric origin, is why QCD and the electroweak force have different characters.

Uniformity of Killing norms. A complete audit of all 21 generators of $\mathfrak{so}(5, 2)$ reveals that every generator has exactly the same Killing norm: $|B(J_{AB}, J_{AB})| = 10$ (negative for compact generators, positive for noncompact). No individual generator is geometrically privileged over any other. The value $\sin^2 \theta_W = 1/3$ emerges entirely from T_3 being a *linear combination* of two generators (J_{01} and J_{23} , Killing norm 20), while $Y_{\text{phys}} = J_{56}$ is a single generator (Killing norm 10). The ratio $1 : (1 + 2) = 1/3$ has no free parameters.

The charged/neutral current split as a theorem. The six color-singlet generators $\{J_{01}, J_{56}, J_{05}, J_{06}, J_{15}, J_{16}\}$ form $\mathfrak{so}(2, 2) \cong \mathfrak{sl}(2, \mathbb{R})_L \oplus \mathfrak{sl}(2, \mathbb{R})_R$. The canonical decomposition, verified to machine precision with explicit generators:

$$\mathfrak{sl}(2, \mathbb{R})_L : \quad h_L = J_{05} + J_{16}, \quad e_L = \frac{1}{2}(J_{01} - J_{06} + J_{15} + J_{56}), \quad f_L = \frac{1}{2}(-J_{01} - J_{06} + J_{15} - J_{56})$$

$$\mathfrak{sl}(2, \mathbb{R})_R : \quad h_R = J_{05} - J_{16}, \quad e_R = \frac{1}{2}(-J_{01} - J_{06} - J_{15} + J_{56}), \quad f_R = \frac{1}{2}(J_{01} - J_{06} - J_{15} - J_{56})$$

with $[h_L, e_L] = 2e_L, [h_L, f_L] = -2f_L, [e_L, f_L] = h_L$ (and likewise for R), and all nine cross-commutators $[\mathfrak{sl}_L, \mathfrak{sl}_R] = 0$ — exact.

The physical interpretation: - $\mathfrak{sl}(2, \mathbb{R})_L$: the charged-current sector — $W^\pm$ bosons; the raising operator e_L carries quantum numbers of the charged weak current - $\mathfrak{sl}(2, \mathbb{R})_R$: the neutral-current sector — Z^0 boson; h_R is the differential boost that becomes the Z ’s longitudinal mode after electroweak symmetry breaking

The charged/neutral current split is not imposed on BST — it is the canonical $\mathfrak{sl}(2, \mathbb{R})_L \oplus \mathfrak{sl}(2, \mathbb{R})_R$ decomposition of the color-singlet subalgebra, forced by the geometry of $\mathfrak{so}(5, 2)$.

Neutrino chirality prediction. Left-handed neutrinos couple to $\mathfrak{sl}(2, \mathbb{R})_L$ (charged current) — they are left-handed because the charged-current algebra is the L factor of the decomposition. If right-handed neutrinos exist, they are color-singlets and $SU(2)_L$ -singlets, so they must couple only to $\mathfrak{sl}(2, \mathbb{R})_R$ (neutral current only, no W coupling). This is a sharp BST prediction: right-handed neutrinos, if they exist, have *no charged-current coupling* — not merely suppressed, but geometrically forbidden by the algebra structure. The search for right-handed neutrino contributions to the charged weak current is therefore a direct test of BST.

Section 7: Nuclear Physics from BST Geometry

7.1 Proton Radius

The proton charge radius is the spatial extent of the Z_3 circuit on $\mathbb{CP}^2$. The circuit extends over all $\dim_{\mathbb{R}}(\mathbb{CP}^2) = 4$ real directions of the circuit space, each contributing one proton Compton wavelength:

$$r_p = \frac{\dim_{\mathbb{R}}(\mathbb{CP}^2)}{m_p} = \frac{4\hbar}{m_p c} = 0.8412 \text{ fm} \quad (0.058\% \text{ from CODATA } 0.84075 \pm 0.00064)$$

The integer $4 = 2(N_c - 1)$ connects the proton's charge radius directly to the color geometry. BST predicts the muonic hydrogen value, siding with the smaller radius in the proton radius puzzle. See `notes/BST_ProtonRadius.md`.

7.2 Nuclear Shell Structure

BST's circuit topology provides a framework for nuclear shell structure. The magic numbers — 2, 8, 20, 28, 50, 82, 126 — correspond to particularly stable circuit configurations on $\mathbb{CP}^2$ with specific topological error correction properties.

The spin-orbit interaction in BST arises from the coupling between a nucleon's circuit winding on S^1 and the angular momentum of its motion on $\mathbb{CP}^2$. The coupling strength is a ratio of BST integers:

$$\kappa_{ls} = \frac{C_2}{n_C} = \frac{6}{5} = 1.200$$

The standard harmonic-oscillator shell model reproduces all seven observed magic numbers once given a spin-orbit strength; BST's contribution is a *consistent factorization* of that strength as $\kappa_{ls} = C_2/n_C = 6/5$ — a real consistency, not a unique

forcing of the magic numbers (the per-number closed forms are post-hoc, per Cal's K601/#286 re-tier). The durable predictive result is the 8th magic number, 184, testable in superheavy element experiments.

BST reduces this nuclear structure calculation to linear algebra: the shell energies are eigenvalues of a finite-dimensional matrix whose entries are Chern class ratios of Q^5 . The spin-orbit matrix element $\kappa_{ls} = C_2/n_C$ is the ratio of the Casimir eigenvalue to the complex dimension — both read directly from the D_{IV}^5 root system. The claim is that the *one strength* the shell model would otherwise fit is factorized into known BST integers; the magic numbers themselves then fall out as eigenvalue crossings of that matrix, exactly as in the standard shell model.

7.3 Confinement — what the geometry gives, and what is imported

The geometry forces the Z_3 winding structure and $N_c = 3$: a quark carries a fractional (third-of- Z_3) winding, and only Z_3 -complete (integer-charge) composites — three quarks, or quark–antiquark — close the circuit and reach the physical boundary. So *which* states are physical is a geometric selection: an isolated fractional-winding quark is not a complete circuit, and (equivalently, at the representation level) a color-nonsinglet two-row K-type has *zero* Shilov boundary value and so cannot be an asymptotic state. This kinematic fact — *which* states can be asymptotic — is derived, and it is why no experiment produces an isolated quark: the geometry forbids it as an asymptotic state.

What the geometry does *not* supply is the SU(3) gauge *dynamics* — the confining force itself (the linear potential, the string tension, the Yang–Mills mass gap). That is imported (the SU(3) gauge principle is not in the geometry's internal algebra; and no geometry-symmetry operator can confine the color triplet). So the honest split is: the geometry says *which* states are physical (Z_3 -complete / color-singlet composites) — a real kinematic result — while the *dynamical* mechanism that binds the quarks is the imported SU(3) gauge principle. The topological picture is the *necessary condition* on asymptotic states, not a derivation of the confining dynamics.

7.4 Proton Mass from Bergman Geometry

$$\boxed{\frac{m_p}{m_e} = (n_C + 1) \pi^{n_C} = 6\pi^5 = 1836.118}$$

vs. observed $m_p/m_e = 1836.153$ — 0.002% agreement with no free parameters.

The formula has two factors, each forced by D_{IV}^5 geometry.

Factor 1: the Bergman kernel power — corrected 2026-09-11. The Bergman kernel of D_{IV}^5 is $K(z, w) \propto N(z, w)^{-p}$ with p the genus, and the genus of the Lie ball in five dimensions is 5 (Xiao–Yuan; pinned to the page 2026-09-09, and checked by Monte Carlo on the domain in Round 130). The May text wrote the power as $n_C + 1 = 6$; that is the *unit ball's* exponent — Elie's toy 5752 (Round 142) measures 6.000

on B^5 and 5.0003 ± 0.0006 on the Lie ball in the same run, with the reproducing property holding at $p = 5$ and failing at $p = 6$ by $60\text{--}80\sigma$ — a rank-one number in a rank-two slot, the same species as T1918. The arithmetic below that used 6 as “the Bergman power” is an identification whose label is now wrong; Grace’s 29-row genus sweep (R140) lists the affected rows. It controls the weight of every mode on the domain and appears throughout the BST structure (Wylar formula for α , fermion mass ratios, Λ derivation). It counts the power of the volume form in the Bergman measure.

Factor 2: $\pi^{n_C} = \pi^5$ — the domain volume factor. This is the geometric volume unit at complex dimension $n_C = 5$: $\pi^5 = n_C! \times 2^{n_C-1} \times \text{Vol}(D_{IV}^5) \times (2^{n_C-1} n_C!)$ in the sense that π^{n_C} encodes the Bergman measure on D_{IV}^5 at full complex dimension.

Physical interpretation. The electron is the minimal S^1 winding: one complete circuit of the simplest topology. The proton is the minimal Z_3 closure: three quarks completing a topological constraint imposed by the Z_3 center of $SU(3)$. The ratio m_p/m_e measures how much more Bergman weight a Z_3 baryon carries compared to a simple winding. At complex dimension $n_C = 5$, this ratio is forced to be $(n_C + 1)\pi^{n_C} = 6\pi^5$.

The 0.002% residual ($\Delta m = 0.034 m_e = 0.017 \text{ MeV}$) is consistent with the electromagnetic self-energy of the proton: the proton is charged (winding number 1 under the $U(1)$ electromagnetic factor), adding a self-energy of order $\alpha \times m_p \times (\text{form factor})$. The formula $6\pi^5$ gives the bare QCD proton mass; the observed mass includes this EM correction.

The neutron-proton mass difference decomposes into QCD and EM contributions, both derived from BST geometry. The total formula $(m_n - m_p)/m_e = 91/36$ (0.13%) gives the EM contribution $\Delta m_{\text{EM}} = -\alpha m_p / \sqrt{30}$ with the $\mathbb{CP}^2$ form factor $F = \sqrt{3/10}$, now derived:

$$F^2 = \frac{\dim_{\mathbb{R}}(\mathbb{CP}^2)}{\dim_{\mathbb{R}}(D_{IV}^5)} \times \frac{\binom{N_c}{2}}{N_c^2} = \frac{4}{10} \times \frac{3}{9} = \frac{3}{10}$$

The first factor is the fraction of the real dimensions occupied by the color fiber $\mathbb{CP}^2$ ($\dim_{\mathbb{R}} = 4$) within the full domain ($\dim_{\mathbb{R}} = 10$). The second factor is the fraction of quark pairings that are distinct ($\binom{3}{2}/3^2 = 3/9 = 1/3$), reflecting the Coulomb sum over quark pairs. The resulting $\Delta m_{\text{EM}} = -\alpha m_p / \sqrt{30} = -1.250 \text{ MeV}$ matches lattice QCD ($-1.00 \pm 0.34 \text{ MeV}$, within 1σ). The same $\sqrt{30} = \sqrt{2N_c n_C}$ appears in the MOND acceleration scale $a_0 = cH_0/\sqrt{30}$, connecting nuclear and galactic physics through the same geometric factor. See notes/BST_NeutronProton_MassSplitting.md.

The Neutron as Shilov Boundary Composite (T958). The Shilov boundary of D_{IV}^5 is $S^4 \times S^1$. The neutron is the physical realization of this product: an unstable

composite that factorizes into irreducible pieces through beta decay. The mapping is exact:

$$n \rightarrow p + e^- + \bar{\nu}_e \quad \longleftrightarrow \quad S^4 \times S^1 \rightarrow S^4 + S^1 + (+1)$$

The proton lives on S^4 (compact, simply connected, topologically stable, color-confined). The electron lives on S^1 (compact, winding-number-protected, light). The antineutrino is the +1 remainder of T914 — the irreducible observer term that appears whenever a composite factorizes at the boundary.

This identification connects to nucleosynthesis: the freeze-out ratio $n/p = 1/7 = 1/g$ at the epoch of primordial nucleosynthesis is the Bergman genus. The universe creates the Shilov boundary (neutrons), and factorization (beta decay) produces the stable particles. Free neutrons are unstable because composites factorize at the boundary; bound neutrons in nuclei are stable because composites embedded in a larger lattice are protected by Pauli blocking — exactly as smooth numbers are stable inside larger smooth products in the T914 framework.

Matter creation is the geometry writing its boundary. Beta decay is the boundary reading itself into components.

7.5 Lepton Mass Spectrum from Bergman Submanifold Embeddings

The three charged lepton generations correspond to totally geodesic submanifolds in the D_{IV}^k tower:

Generation	Domain	$\dim_{\mathbb{C}}$	$\dim_{\mathbb{R}}$
Electron	D_{IV}^1	1	2
Muon	D_{IV}^3	3	6
Tau	D_{IV}^5	5	10

with the embedding hierarchy $D_{IV}^1 \subset D_{IV}^3 \subset D_{IV}^5$. The domain volumes follow the exact formula $\text{Vol}(D_{IV}^k) = \pi^k / (2^{k-1} k!)$:

Domain	Volume	Bergman kernel $K_k(0, 0)$
D_{IV}^1	π	$1/\pi$
D_{IV}^3	$\pi^3/24$	$24/\pi^3$
D_{IV}^5	$\pi^5/1920$	$1920/\pi^5$

Muon-electron mass ratio.

$$\frac{m_\mu}{m_e} = \left[\frac{K_3(0, 0)}{K_1(0, 0)} \right]^{\dim_{\mathbb{R}}(D_{IV}^3)} = \left(\frac{24}{\pi^2} \right)^6 = 206.761$$

vs. observed $m_\mu/m_e = 206.768$ — 0.003% agreement, no free parameters.

The base $24/\pi^2 = K_3(0,0)/K_1(0,0)$ is the ratio of Bergman kernels between the muon domain and the electron domain. The exponent $6 = \dim_{\mathbb{R}}(D_{IV}^3)$ is the real dimension of the muon submanifold — it enters because the mass ratio is a real (not holomorphic) invariant, requiring integration over the full real fiber bundle of D_{IV}^3 . Geometrically, $(24/\pi^2)^6$ is the determinant of the Bergman curvature transformation — the full Jacobian of the real embedding $D_{IV}^1 \hookrightarrow D_{IV}^3$.

Equivalently: $m_\mu/m_e = \exp(\dim_{\mathbb{R}}(D_{IV}^3) \cdot \Delta S_{\text{Bergman}})$, where $\Delta S_{\text{Bergman}} = \ln K_3(0,0) - \ln K_1(0,0)$ is the Bergman entropy difference between generations.

Tau and quark mass relations.

Ratio	BST formula	BST value	Observed	Error
m_μ/m_e	$(24/\pi^2)^6$	206.761	206.768	0.003%
m_τ/m_e	$(24/\pi^2)^6 \times (7/3)^{10/3}$	3483.8	3477.2	0.19%
m_τ/m_μ	$(7/3)^{10/3} = (\kappa_1/\kappa_5)^{2n_C/N_C}$	16.850	16.817	0.19%
m_t/m_c	$N_{\max} - 1$	136	135.98	0.017%*
m_s/m_d	$4n_C$	20	$20.0 \pm \sim 5\%$	$\sim 0\%$
m_b/m_τ	$g/N_C = 7/3$ ($g = p+q = 7$, the defining- representation dimension, not the genus)	2.333	2.352	0.81%
m_b/m_c	$\dim_{\mathbb{R}}(D_{IV}^5)/N_C = 3.333$ 10/3		3.291	1.3%*
m_c/m_s	$N_{\max}/\dim_{\mathbb{R}} = 137/10$	13.7	13.6	0.75%

*Within experimental uncertainty on m_t and heavy quark masses.

The lepton mass hierarchy has a two-step geometric structure. Step 1 (electron $\rightarrow$ muon): the volume Jacobian of $D_{IV}^1 \hookrightarrow D_{IV}^3$, giving $(24/\pi^2)^6$. Step 2 (muon $\rightarrow$ tau): the holomorphic sectional curvature ratio $(7/3)^{10/3}$, where $7/3 = \kappa_1/\kappa_5 = \text{genus}/N_C$ is the curvature ratio and $10/3 = 2n_C/N_C = \dim_{\mathbb{R}}(D_{IV}^5)/N_C$ is the real dimension per color. In physical units: $m_\tau(\text{BST}) = 1780.2$ MeV vs. observed 1776.86 ± 0.12 MeV. The exponent derivation for the muon (proving that $\dim_{\mathbb{R}}$ is the correct power from embedding theory) is also open.

Full derivation and numerical verification: `notes/BST_FermionMass.md`, `notes/BST_ProtonMass.md`.

Quark mass ratios from D_{IV}^5 invariants. The quark mass hierarchy is organized by the same geometric invariants. Six quark mass ratios are identified as clean D_{IV}^5 numbers:

- $m_s/m_d = 4n_C = 20$ (exact to measurement precision). The same $4n_C$ that appears in $\sin^2 \theta_C = 1/(4n_C) = 1/20$ — CKM mixing and quark masses share a common geometric origin.
- $m_t/m_c = N_{\max} - 1 = 136$ (0.017%). The top saturates the vacuum minus one level.
- $m_b/m_\tau = g/N_c = 7/3$ (0.81%; $g = p + q = 7$ is the defining-representation dimension of $SO(5, 2)$, not the genus). Third-generation quark-lepton partners coupled by the holomorphic curvature ratio $\kappa_1/\kappa_5 = 7/3$.
- $m_b/m_c = \dim_{\mathbb{R}}(D_{IV}^5)/N_c = 10/3$ (1.3%). The real dimension per color.
- $m_c/m_s = N_{\max}/\dim_{\mathbb{R}} = 137/10$ (0.75%). Bridging thermal and geometric sectors.

The third generation is universally stamped by $N_c = 3$ as denominator: $m_b/m_\tau = 7/3$, $m_b/m_c = 10/3$, m_τ/m_μ exponent = $10/3$. Color is the denominator of the third generation because it probes the full D_{IV}^5 domain where all N_c color channels are active. Full details: `notes/BST_QuarkMassRatios.md`.

7.6 Neutrino Masses from the Boundary Seesaw

The three neutrino mass eigenstates are derived from the boundary seesaw: the BST analog of the standard seesaw mechanism, where the Dirac mass is m_e (boundary excitation) and the Majorana mass is m_p (bulk excitation).

$$m_{\nu_i} = f_i \times \alpha^2 \times \frac{m_e^2}{m_p}$$

The seesaw base $\alpha^2 \times m_e^2/m_p = 0.01482$ eV involves two factors of α from two electroweak vertices on the contact graph (the neutrino couples to the Hopf fiber, which couples back to the boundary mass sector). The geometric factors f_i encode coupling to the D_{IV}^5 geometry:

Neutrino	Geometric factor f_i	BST mass (eV)	Observed (eV)	Deviation
ν_1	0	0	< 0.009	—
ν_2	$(n_C + 2)/(4N_c) = 7/12$	0.00865	≈ 0.00868	−0.35%
ν_3	$2n_C/N_c = 10/3$	0.04940	≈ 0.0503	−1.8%

The neutrino as vacuum quantum. The massless ν_1 is not merely light — it IS the

vacuum ground state of D_{IV}^5 . The lightest neutrino is a pure Goldstone mode of the broken Z_3 symmetry at the Shilov boundary, with mass forbidden by residual Z_3 symmetry that protects baryon number. Neutrino oscillation is the vacuum shifting between its three geometric modes. The heavier neutrinos ν_2 and ν_3 are vacuum fluctuations — excitations of the vacuum quantum.

This identification resolves the cosmic coincidence problem: $\Lambda^{1/4} \sim m_\nu$ because both scale as $\alpha^{14} m_{\text{Pl}}$ with $14 = 2(n_C + 2) = 2 \times \text{genus}$, and $\Lambda \sim \alpha^{56} = (\alpha^{14})^4 \propto m_\nu^4$. The “coincidence” is a geometric identity.

The mass ratio $m_3/m_2 = 40/7 = 5.714$ is a pure D_{IV}^5 geometric ratio depending only on $n_C = 5$. BST predicts normal ordering with $m_1 = 0$ exactly and $\Sigma m_\nu = 0.058$ eV.

Full derivation: `notes/BST_NeutrinoMasses.md`. Vacuum quantum connection: `notes/BST_VacuumQuantum_NeutrinoLambda.md`.

7.7 CKM and PMNS Mixing

Rewritten 2026-09-13 (Lyra, Round 143 L2; Keeper audit pending) as a pointer into the Spine plus the referee apparatus. The May 2026 tables and the K1801/K1809 curation flag of 2026-08-22 are kept below as the record. Facts are from the register (T2519, T2529, T2530, T2532, T2534, T2536, T2547; K1635, K1799, K1800, K1801, K1808, K1809, K1810) and from data/bst_26_tier_table_generated.md rows 10–17 (Grace’s generator, keyed to Cal Section 946) — not from the May text. No tier table is typed here; the generated one is the table. Last accuracy-synced: 2026-09-13 / K1895.

7.7.0 The question Why do quarks — and neutrinos — of different generations mix the way they do, and what exactly did the geometry fix before anyone looked?

The Standard Model takes all four quark-mixing parameters and all four lepton-mixing parameters as inputs. The quark half is Lecture 7 of the Spine (`Curriculum/Spine_DIV5_QM_GR_SM/Lecture_07_Mixing.md`); this section is the pointer to it, the lepton half as the register carries it, and the apparatus.

7.7.1 Where it stands Quarks. Generation space is three-dimensional (Lecture 6), and the geometry fixes the *skeleton* of the mixing: a rank-one alignment on the leading stratum, so the matrix is nearly the identity (T2519); flavour universality is the partial-isometry condition, and given unitarity the only survivor is $V = U_{\text{up}}^\dagger U_{\text{down}}$. Two mechanisms are dead and usefully so: mixing is not radial (K1635), and it is not carried by the parity fold of the quadric, which is a projector with spectrum $\{0, 1\}$ and no scale of its own (K1799). One value is derived, blind: on the down-type module the texture has a zero in the (1,1) place, diagonalising it gives the Gatto relation $\tan \theta_C = \sqrt{m_d/m_s}$, and with $m_s/m_d = \text{rank}^2 n_C = 20$ (T2529) the Cabibbo angle is $\lambda = 1/\sqrt{20} = 0.22361$, dated 2026-07-29 before the comparison (T2530). Against PDG 2024’s direct value 0.22501 ± 0.00068 it sits at -2.06σ ; with $|V_{ud}| = \sqrt{19/20}$ the first row is exactly unitary, where the direct determinations

sum to 0.9984 ± 0.0007 , 2.3σ short — a tension the experiments report themselves. Which side we score against is pre-registered, not chosen after (Lecture 7). The May form $2/\sqrt{79}$, at $+0.01\sigma$, is retired: it fit better and was tunable, $1/\sqrt{20}$ was blind (the T1444 form retired 2026-08-23; a competitor *count* is target-independent, a ranking is not). One order is derived with no knob in it: on a three-dimensional generation space Cayley–Hamilton collapses every series in the even part S of the squared mixing generator to $\beta S + \alpha S^2 + \gamma$, with $S_{13} = 0$ and $(S^2)_{13} = 1$, so the 1–3 corner opens two rungs after the 2–3 subdiagonal — one power of the expansion parameter t later (K1808, corrected and confirmed K1810). The corner’s *value*, $t/(1+4t)$ with $t \in [0.120, 0.190]$, is input: five candidate series were sealed by hash before scoring and all five missed (K1800; the register’s E6). CP violation *exists* by a theorem — a real domain has no natural phase and a complex one does; the up and down sectors are different mechanisms (T2547) — and its magnitude is open: $\gamma = \arctan \sqrt{n_C} = 65.9^\circ$ admits ten competing integer forms inside the PDG band and is not cited as derived (K1809). A is input (the May 4/5 is a candidate with no mechanism, -1.7σ against 0.826 ± 0.015); $|V_{cb}|$ ’s *position* $A\lambda^2$ is kept and its May value 0.044 is retired at $+9.2\sigma$ against the exclusive $(39.77 \pm 0.46) \times 10^{-3}$, with the inclusive/exclusive split of more than 3σ noted rather than exploited (the generator prints 8.9σ on its own pin; the two are one retirement, and Grace owns the reconciliation).

Leptons. The register, as of 2026-09-11, carries the four PMNS parameters at three different words, and this section says which. $\sin^2 \theta_{23}$ is derived as maximal, $1/2$, by two independent proofs — the Šilov $\mathbb{Z}_2$ and the parity theorem (T2534); the departure from maximal is *not* claimed, and the May form 4/7 is an identification of that departure (against NuFIT 6.0’s normal-ordering best fit near 0.47, in the lower octant, the May agreement with 0.572 was with an older fit). $\sin^2 \theta_{13} = 1/45$ is derived on the register’s ladder with an owed re-read — the value is pinned and falsifiable (0.02222 against 0.02195), the mechanism is not yet stated in the four-word form, and the K1809 sibling test has not been run on it. $\sin^2 \theta_{12} = 3/10$ is identified (2% against 0.307; the corpus spells the same number two ways, $N_c/(2n_C)$ and $N_c/(N_c + g)$, which is itself the K1809 species). The CP phase’s *existence* is forced by the same mechanism as the quarks’ (T2536); its magnitude is off — the May $12\pi/7 = 308.6^\circ$ is a 4.5σ miss against T2K/NOvA’s $195^\circ \pm 25^\circ$ (K1801 finding 5; K1304) — and the branch is data-picked. The May text’s “PMNS angles are large because neutrinos are vacuum modes” is a reading, not a row. What the lepton sector does *not* have is a Lecture: the Spine carries it only through Lecture 6’s strata and Lecture 5’s ν_R mechanism, and a reader who wants the PMNS derivations goes to the register rows named above.

7.7.2 Apparatus — kept, labelled as apparatus The May tables below are the apparatus a referee will want: every formula the program ever wrote for these eight parameters, with the measured values it compared against on the day. They are not the current tier table; that is generated, never typed — `data/bst_26_tier_table_generated.md`, rows 10–13 (CKM) and 14–17 (PMNS),

keyed to Cal Section 946 — and `python3 play/bst_26_tier_generator.py --emit` rebuilds it from the key file. The K1801/K1809 curation flag is retained at the head of the record because it is the audit that found what the tables claimed; its seven findings are what 7.7.1 applies. The Cayley–Hamilton apparatus (the six-rung path, $S = Q^2|_{\text{even}}$, characteristic polynomial $\lambda^3 - 5\lambda^2 + 6\lambda - 1$, the corner ratio $t/(1 + 4t)$) is in K1808 and K1810; the five sealed series and their hash are K1800; the ten competitors of γ are enumerated in K1809.

7.7.3 Tier line

- Derived: the skeleton (T2519); flavour universality as the partial-isometry condition; $\lambda = 1/\sqrt{20}$, blind (T2530 on T2529); CP existence, both sectors (T2547, T2536); the *order* of the corner (K1808/K1810); θ_{23} maximal (T2534); $\theta_{13} = 1/45$ on the ladder, re-read owed.
- Identified: $\theta_{12} = 3/10$; the 4/7 departure of θ_{23} ; $|V_{cb}|$'s position.
- Fired and lost, certified: five sealed series for the corner's value (K1800/E6).
- Open (the register's word; “input” is reserved for the two dimensionful scales): the corner's value; δ_{CP} 's magnitude; A — taken from measurement where used.
- Open: γ , $\bar{\rho}$, $\bar{\eta}$, J (K1809: report the count, not the winner; $J = \sqrt{2}/50000$ is untested, not passed); δ_{PMNS} 's magnitude.
- Live falsifiers: first-row CKM unitarity resolving against unity (A-row of the register); the corner's order read across $|V_{us}| : |V_{cb}| : |V_{ub}|$ (A8, owed to Grace this round).
- Not claimed: exact values beyond λ ; any “smallest of N ” without N ; “the full CKM and PMNS matrices derived” (K1801 finding 6).

7.7.4 May 2026 record *Below: the K1801/K1809 curation flag of 2026-08-22 and the May tables, with the in-place corrections of 2026-08-23 that the flag's clearance applied to the CKM rows (λ , A , $|V_{cb}|$, $|V_{ub}|$). Unedited on 2026-09-13; read with 7.7.1 in hand.*

🔍 KEEPER CURATION FLAG — K1801, 2026-08-22. This section claims more than the research layer proves. Six findings, all measured this round. Nothing below is rewritten pending Grace (ledger), Lyra (T1444/T2530), Cal (cold-read) and Casey (GO). Do not dispatch this section until they clear.

1. Three mutually inconsistent values of λ appear in this one section. The Cabibbo row uses $2/\sqrt{79} = 0.225018$ (T1444). Our banked, Derived value is $\lambda = 1/\sqrt{20} = 0.223607$ (T2530, pre-dated blind) — and Section 7.4 of *this same chapter* states $\sin^2 \theta_C = 1/(4n_C) = 1/20$. The $|V_{cb}|$ closed form implies a third value, $\lambda = 1/5$. Spread: +0.63% and −10.56% against the banked value.
2. The $|V_{cb}|$ row is internally inconsistent. The stated closed form $4/125 = 0.0320$; the value column says 0.0400. 20% apart. With

$A = 4/5$ and the banked $\lambda^2 = 1/20$ the correct closed form is $4/100 = 1/25$. Not corrected here because the fix depends on resolving finding 1.

3. A is presented as derived; in the research ledger it is the open input. The Guide gives $A = (n_C - 1)/n_C = 4/5$. The current CKM ledger carries $c_{cb} = 0.816$ as INPUT, not derived (Partially Derived, explicit-split). Against PDG 0.825 ± 0.012 the Guide's 0.800 is -2.1σ . A clean rational form for an open parameter is a candidate until a mechanism is exhibited.

4. The experimental column is stale — and correcting it helps BST here. Guide quotes PDG 2024 $|V_{cb}| = 0.0411 \pm 0.0013$. Current exclusive is $(39.77 \pm 0.46) \times 10^{-3}$, with the inclusive/exclusive split now $> 3\sigma$ and unresolved — unmentioned in this section. Against current exclusive, BST's 0.0400 sits at $+0.5\sigma$, materially *better* than the -2.7% this table claims against stale data. The $|V_{ub}|$ error bar quoted here (± 0.000110) is roughly $3\times$ tighter than current Belle. Which side of the $> 3\sigma$ split we score against must be pre-registered, not chosen.

5. The PMNS δ_{CP} row hides a large miss behind a soft phrase. BST $12\pi/7 = 308.6^\circ$ against T2K/NOvA $195^\circ \pm 25^\circ$ is a 4.5σ miss (the -51.4° wrap-around is worse, 9.9σ). The deviation column reads “*measurement evolving*” and gives no number. Every other row in these tables carries a numeric deviation; this one must too.

6. Related over-claim downstream. Vol6_Frontier/Ch01_Deep_Results.md lists “the full CKM and PMNS matrices” among derived results. The CKM sector is Partially Derived, 1-of-4 explicit-split (λ Derived; c_{cb} , c_{ub} , δ_{CP} open), with the V_{cb} coarse bank itself under re-score (K1800h).

7. ADDED K1809, 2026-08-22 — the T2198 retirement was never swept, and three rows here fail it. T2198/T2259 were retired because “*the band admits four consecutive integers, all indistinguishable.*” The same test was never run on the five sibling formulas below. Run now: $\gamma = \arctan \sqrt{n_C}$ admits TEN competing BST-integer forms inside $65.5 \pm 2.5^\circ$; $\bar{\eta} = 1/(2\sqrt{2})$ admits FOUR; $\bar{\rho} = 1/(2\sqrt{2n_C})$ admits TWO. And the acute case: for $\bar{\eta}$ a competing BST form fits BETTER than the published one — $\sqrt{C_2}/g = \sqrt{6}/7 = 0.34993$ at $+0.09\sigma$ against the published 0.35355 at $+0.46\sigma$, same vocabulary, same simplicity, $\sim 5\times$ closer. A referee needs only a calculator to find this. $J_{CKM} = \sqrt{2}/50000$ is UNTESTED, not passed — the sweep's pool generates simple forms and cannot reach its compound class ($50000 = n_C^5 (2^{\text{rank}})^2$). Action: retire the three failing rows, or re-state them as “smallest-of-N indistinguishable BST forms” with N reported. This touches none of the sector's genuine results — $\lambda = 1/\sqrt{20}$, the derived mixing ORDER, flavor-universality, and CP existence are unaffected.

— Keeper K1801 + K1809. Filed, not fixed: these are physics decisions, not typos. The tables stand as written so the record is legible; they must not ship in this state.

The quark and lepton mixing matrices encode the mismatch between mass eigenstates (Bergman bulk modes, from H_{YM}) and weak-interaction eigenstates (Hopf fiber modes, from $S^3 \rightarrow S^2$ geometry). All six mixing angles are ratios of $n_C = 5$ and $N_c = 3$.

PMNS (neutrino mixing) — large angles:

Angle	BST formula	BST value	NuFIT 5.3	Deviation
$\sin^2 \theta_{12}$	$N_c/(2n_C) = 3/10$	0.300	0.303 ± 0.012	−1.0%
$\sin^2 \theta_{23}$	$(n_C - 1)/(n_C + 2) = 4/7$	0.5714	0.572 ± 0.018	−0.1%
$\sin^2 \theta_{13}$	$1/(n_C(2n_C - 1)) = 1/45$	0.02222	0.02203 ± 0.00056	+0.9%
δ_{CP}	$12\pi/7 = 2\pi(C_2/g)$	308.6°	$195^\circ \pm 25^\circ$ (T2K/NOvA)	measurement evolving

Note on PMNS derivation routes. The formulas above use representation-theoretic ratios of n_C and N_c — the same rational-function pattern as CKM. An alternative route via tribimaximal mixing gives $\{\sin^2 \theta_{12} = 1/3, \sin^2 \theta_{23} = 1/2, \sin^2 \theta_{13} = 3/137\}$, which is better for θ_{13} (closer to $3/N_{\max}$) but significantly worse for θ_{12} (2σ deviation) and θ_{23} (4σ). The representation-ratio formulas are retained as primary because they match all three angles simultaneously and arise naturally from the D_{IV}^5 boundary geometry.

CKM (quark mixing) — small angles:

Parameter	BST formula	BST value	PDG 2024	Deviation
$\lambda = \sin \theta_C$ (Cabibbo)	$1/\sqrt{20}$ — the Gatto relation $\tan \theta_C =$ $\sqrt{m_d/m_s}$ is the geometry; DERIVED, blind, 2026-07-29 (T2530). The May form $2/\sqrt{79}$ (T1444) is retired.	0.22361	$0.22501 \pm$ 0.00068	-2.06σ on the direct value; exactly unitary in the first row, where the direct deter- minations sum to $0.9984 \pm$ 0.0007 — which side we predict is pre- registered, not chosen 4/5 would sit at -1.73σ
A (Wolfenstein)	INPUT (the May form 4/5 is a candidate with no mechanism; K1801 finding 3)	—	0.826 ± 0.015 (PDG 2024 Rev 12; re-pinned 2026-08-23)	
$\ V_{cb}\ $	$A\lambda^2$; IDENTIFIED (the register's word; position $A\lambda^2$, value input) (the old 0.044 retired at $+9.2\sigma$ vs current exclusive ($39.77 \pm$ $0.46) \times 10^{-3}$; the inclu- sive/exclusive split is $> 3\sigma$ and unresolved)	—	see left	—

Parameter	BST formula	BST value	PDG 2024	Deviation
$\ V_{ub}\ $	ORDER DERIVED: exactly one power of λ below $\ V_{cb}\ $ (Cayley- Hamilton on the 3- dimensional generation space; the 1-3 corner opens two rungs after the 2-3; K1808/K1810, zero-knob). VALUE INPUT: five named series sealed by hash before scoring, all five missed.	—	PDG 2024 Rev 12 (pinned there, not retyped here)	—

CKM CP violation — Wolfenstein parameters $\bar{\rho}$, $\bar{\eta}$, and the unitarity triangle:

Parameter	BST formula	BST value	PDG 2024	Deviation
γ (CP phase)	$\arctan(\sqrt{n_C}) =$ $\arctan(\sqrt{5})$	65.91°	$65.5^\circ \pm 2.5^\circ$	0.6%
$\bar{\rho}$	$1/(2\sqrt{2}n_C) =$ $1/(2\sqrt{10})$	0.158	0.159 ± 0.010	0.6%
$\bar{\eta}$	$1/(2\sqrt{2})$	0.354	0.349 ± 0.010	1.3%
J_{CKM} (Jarlskog)	$\sqrt{2}/50000$	2.83×10^{-5}	$(2.77 \pm$ $0.11) \times 10^{-5}$	2.1%

Structural relations: - $\bar{\eta}/\bar{\rho} = \sqrt{n_C}$ exactly: the ratio of the two Wolfenstein parameters is the square root of the domain dimension. - $\sin^2 \gamma = n_C/(n_C + 1) = 5/6$: the CP phase follows the same rational-function-of- n_C pattern as all other mixing parameters. - $R_b = \lambda\sqrt{N_c} = \sqrt{3/20}$: the unitarity triangle base is Cabibbo $\times \sqrt{\text{colors}}$. - $J_{\text{CKM}} = \sqrt{2}/50000$ where $50000 = n_C^5 \times (2^{\text{rank}})^2 = 3125 \times 16$: the Jarlskog invariant denominator is the domain dimension to the 5th power times the square

of the rank exponential. - $|V_{ub}| = A\lambda^3/\sqrt{C_2} = 1/(50\sqrt{30})$: the $\sqrt{C_2}$ suppression in the third-generation mixing arises from the Casimir cost of the b -quark Bergman embedding.

Physical insight: PMNS angles are large because neutrinos are vacuum modes — they rotate freely on the Shilov boundary with no Bergman embedding cost. CKM angles are small because quarks carry full Bergman weight in the bulk of D_{IV}^5 — the overlap between mass and weak eigenstates is suppressed by the Bergman embedding cost. The CP-violating phase $\gamma = \arctan(\sqrt{n_C})$ encodes the geometric angle between the Bergman bulk and the Shilov boundary in the CKM sector.

Full derivation: `notes/BST_CKM_PMNS_MixingMatrices.md`.

Section 8: Hadronic Spectrum Estimates

8.1 Bare Geometric Values

BST geometric calculations give estimates for hadronic quantities that are systematically below observed values:

Quantity	BST bare	Observed	Ratio
Pion mass	25.6 MeV	140 MeV	$\sqrt{30} = 5.477$
String tension	0.061 GeV ²	0.18 GeV ²	3.0
Glueball mass	490 MeV	1.5–1.7 GeV	3.1–3.5
$g_{\pi NN}$ coupling	3.4	13.5	4.0

The discrepancies are systematic and traceable to a single physical effect: the chiral condensate.

8.2 Distinction: Predictions vs. Estimates

It is important to distinguish between BST *predictions* — quantities derived from the domain geometry with no adjustable parameters — and BST *estimates* — quantities that require additional physics (the chiral condensate) to be quantitative. The parameter-free predictions (Table in Section 43) are strong results. The hadronic estimates are order-of-magnitude geometric calculations that require condensate corrections.

8.3 Vector Meson Masses: ρ and ω (March 2026)

The proton mass uses $C_2 = n_C + 1 = 6$ Casimir eigenvalue units — n_C complex dimensions plus one extra from the Z_3 circuit closure. A meson ($q\bar{q}$) has no Z_3 closure; the quark and antiquark share the same color space. The meson mass formula uses n_C slots instead of C_2 :

$$m_\rho = n_C \times \pi^{n_C} \times m_e = 5\pi^5 m_e = 781.9 \text{ MeV}$$

$$\frac{m_\rho}{m_p} = \frac{n_C}{C_2} = \frac{n_C}{n_C + 1} = \frac{5}{6}$$

Observed: $m_\rho = 775.26 \pm 0.25 \text{ MeV}$ (0.86% error). The isoscalar partner $\omega(782)$ is the same geometric object in a different isospin state: $m_\omega(\text{BST}) = 5\pi^5 m_e = 781.9 \text{ MeV}$ vs. observed $782.66 \pm 0.13 \text{ MeV}$ (0.10% error).

The ratio $m_\rho/m_p = 5/6$ is a structural constant of D_{IV}^5 : a meson is 5/6 of a baryon because it needs one fewer geometric dimension.

The $\phi(1020)$ meson ($s\bar{s}$) uses $(N_c + 2n_C)/2 = 13/2$ slots — the strange quark probes the full color-plus-weak structure:

$$m_\phi = \frac{13}{2} \pi^5 m_e = 1016.4 \text{ MeV} \quad (0.30\% \text{ from observed } 1019.5 \text{ MeV})$$

The $K^*(892)$ ($q\bar{s}$, one light and one strange quark) is the geometric mean:

$$m_{K^*} = \sqrt{n_C \times \frac{N_c + 2n_C}{2}} \pi^5 m_e = \sqrt{\frac{65}{2}} \pi^5 m_e = 891.5 \text{ MeV} \quad (0.02\% \text{ from observed } 891.7 \text{ MeV})$$

The geometric mean is 80× more accurate than the Gell-Mann–Okubo formula ($m_{K^*}^2 = (m_\rho^2 + m_\phi^2)/2$, which gives 1.7% error). This reflects BST's multiplicative mass structure: meson mass factors are eigenvalues of Bergman kernel restrictions, which combine as products (geometric means), not sums (arithmetic means).

8.4 Baryon Resonance Spectrum

If the 1920 cancellation is a property of the domain D_{IV}^5 (via its Weyl group $\Gamma = S_5 \times (\mathbb{Z}_2)^4$, $|\Gamma| = 1920$), independent of the representation, then the baryon mass formula generalizes:

$$m(k) = C_2(\pi_k) \times \pi^{n_C} \times m_e = k(k-5) \times \pi^5 \times m_e, \quad k \geq 6$$

k	$C_2 = k(k-5)$	Mass (MeV)	Candidate	Parity $(-1)^k$
5	0	0	vacuum	—
6	6	938.3	proton (938.3)	+
7	14	2189	N(2190) G_{17} (4★)	—
8	24	3753	prediction	+

The $k = 7$ prediction matches the N(2190) resonance; $k = 8$ at 3753 MeV with positive parity is an open prediction. The $SO(3)$ embedding in $SO(5)$ is resolved: the physical rotation group embeds via the irreducible D_2 representation (the 5 complex dimensions transform as the traceless symmetric tensor $\text{Sym}_0^2(\mathbb{R}^3)$). This gives $L_{\max} = 2N$ at excitation level N , yielding $J^P = 7/2^-$ at $k = 7$ (matching N(2190)) and $J^P \leq 11/2^+$ at $k = 8$. See notes/BST_BaryonResonances_MesonMasses.md.

8.5 Pion Charge Radius via VMD

Using vector meson dominance with BST-derived m_ρ at NLO (VMD + ChPT two-pion loop corrections):

$$r_\pi = \frac{\sqrt{6}}{m_\rho} \Big|_{\text{NLO}} = 0.656 \text{ fm}$$

Observed: $0.659 \pm 0.004 \text{ fm}$ (0.5% error). The NLO VMD + ChPT calculation closes the leading-order gap from 6.2% to 0.5% (Miss Hunt Day, April 9, 2026).

Section 9: Speed of Light and Special Relativity

9.1 Why c Is Constant

If emergent distance = number of bubble contacts, and emergent time = number of causal steps, and each contact IS one step, then distance/time = 1 contact/1 step. Always. In every frame. The speed of light is constant because it is the ratio of a thing to itself, measured in different emergent units.

9.2 Time Dilation

Every bubble has a fixed causal budget per unit of causal time. At rest, all budget goes to internal state changes (aging, ticking, chemistry). In motion, some budget is spent on spatial displacement. The fraction left for internal processing is $\sqrt{1 - v^2/c^2}$ — the Lorentz factor, derived from the Pythagorean structure of the causal budget.

A moving clock ticks slower because its bubbles are busy moving. Photons don't experience time because their entire budget is consumed by spatial propagation.

9.3 $E = mc^2$

Mass is the density of circuit winding on S^1 . Energy is the rate of causal processing. The equivalence $E = mc^2$ follows from c being the universal conversion factor between spatial and causal distance on the contact graph. Since $c = 1$ in natural units (one contact per step), $E = m$ — energy and mass are the same quantity measured in the same units. The factor c^2 in SI units is the unit conversion between meters and seconds, squared.
